

C Comparison to the NIAH test

We extend the analysis of Section 3.2 on the comparison between the task design of `AbsenceBench` and the NIAH test. In table 10, we show 3 LLMs achieve near perfect results in the NIAH test under the poetry domain, as well as substantially improved results within the GitHub PRs domain.

Models	Poetry	Poetry (NIAH)	GitHub PRs	GitHub PRs (NIAH)
Claude-3.7-Sonnet	73.5	99.52	35.7	97.1
GPT-4.1-mini	30.2	99.5	31.3	92.1
Llama-4-Maverick	32.8	99.47	29.0	86.2

Table 10: Micro F1-score (%) of three LLMs evaluated on the `AbsenceBench` and the NIAH test setting under the Poetry and GitHub PRs domains.

D Additional Results

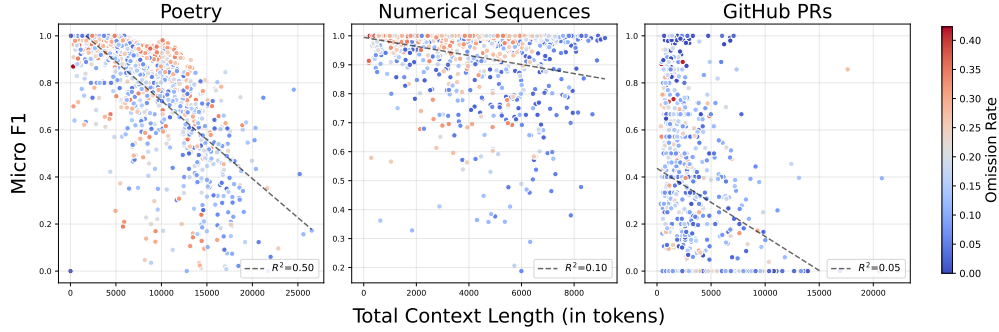


Figure 6: Micro-F1 score of **Claude-3.7-Sonnet** (y-axis) as a function of the total context length (x-axis) as well as the percentage of omission (color)

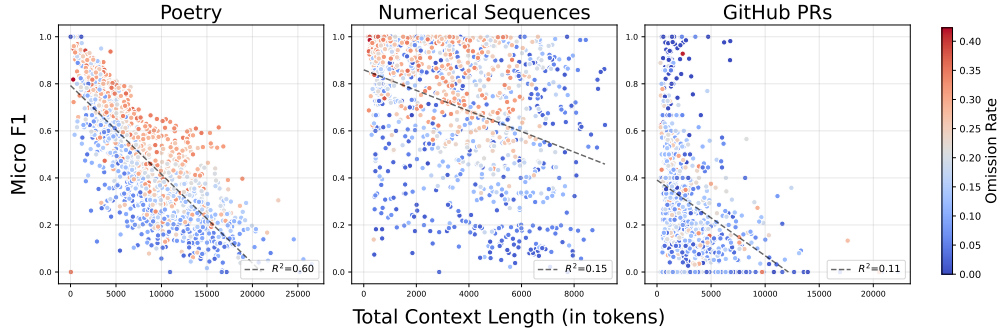


Figure 7: Micro-F1 score of **Llama-4-Maverick** (y-axis) as a function of the total context length (x-axis) as well as the percentage of omission (color)

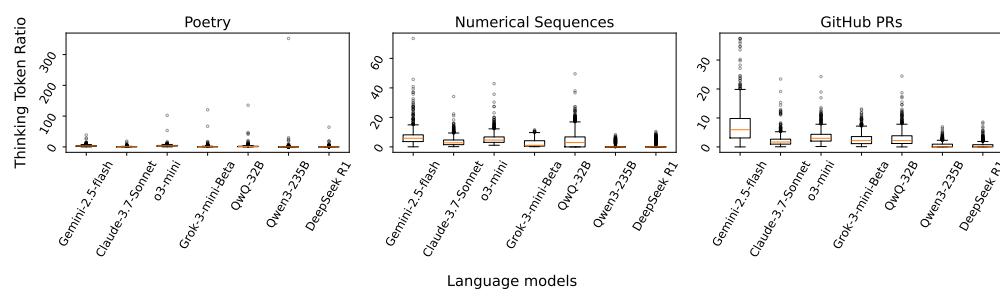


Figure 8: The thinking token ratio (number of generated thinking tokens divided by number of tokens in the original context) for four inference-time compute models under each domain.